



Product Analytics Case Study

Lead-Source Cohort Performance, Funnel Analysis &
FT Conversion Drivers

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Prepared using Excel, SQL (BigQuery Standard SQL) and
Python (scikit-learn) for the driver model

1. Executive Summary

Business problem: Vahan sources gig-work candidate leads from many channels ('lead sources' / cohorts), calls them, and tracks whether they eventually go First Trip (FT) the outcome that matters. This case asks which cohorts convert best, what an aggregated view of the data looks like, and what factors predict FT.

What I analyzed: 18,198 lead-level rows across 16 cohorts uploaded between 18 Jul and 6 Aug 2026. Every row is one lead: one candidate_phone uploaded once into one cohort on one date. The full funnel Lead → Attempted → Connected → Interested → OB → FT was rebuilt from the underlying 0/1 flags, cross-checked in Excel and SQL.

Which cohorts perform best: Ranked on Lead → FT_after_upload conversion, with a minimum size of 1,000 leads and a minimum age of 5 days since upload (to avoid rewarding cohorts that simply haven't had time to convert yet), the **top 3** are:

- Single Referral > 7 days - 24th Jul: 0.933% (14 / 1,500 leads)
- Khanna - 2W 26th Jul: 0.906% (14 / 1,546 leads)
- PreOb-Ob Fees Paid 29th Jul (set 1): 0.472% (7 / 1,483 leads)

What drives FT conversion: a logistic regression (5-fold cross-validated, class-weighted for the 0.3% base rate) shows the single strongest driver is simply whether a lead was ever attempted at all more predictive than any cohort effect. Certain cohorts (Single Referral, Khanna 2W, both PreOb-Ob batches) retain a positive, cohort-specific lift even after controlling for attempt/connect behavior. Model discrimination is reasonable (ROC-AUC 0.80) but precision at any fixed threshold is low, which is expected at this event rate (54 FT events total) the model is fit for ranking/prioritization, not hard pass/fail decisions.

Biggest funnel opportunity: over a third of all leads (34.2%) are never attempted even once the largest single leak in the funnel, and the one most within Vahan's direct operational control. Of the leads that are attempted, more than half never connect.

What Vahan should do **next**:

- (1) close the 34% never-attempted gap first, since it requires no new acquisition spend to unlock;
- (2) shift lead-sourcing mix toward the Single Referral and Khanna-2W-style channels, whose FT edge holds up under the driver model, not just in raw counts;
- (3) treat 'Interested' as a coaching/process metric, not a funnel gate, since the data shows most OB/FT candidates never receive that tag;
- (4) revisit the >19-day-old, zero-FT 2W3W batches their underperformance is not a data artifact.

2. Problem Understanding

The case brief asks three explicit questions:

- Which 3 cohorts are the best, and on what metric?
- Aggregate the raw data at an appropriate level via a SQL query, and provide the aggregated output.
- Build a model that shows which factors influence the chance of FT, including a confusion matrix.

This report answers all three, plus the supporting data-quality, funnel, and recommendation work needed to defend the answers in an interview.

3. Dataset & Analytical Grain

The workbook contains two tabs: a case-brief tab, and a Raw Data tab of 18,198 rows and 19 columns.

Profiling results:

- Grain: one row = one lead. The 'Uploaded Leads' column is 1 on every single row, confirming this directly.
- 16 distinct cohorts (lead_source), 11 distinct upload_date values, spanning 18 Jul - 6 Aug 2026.
- Every cohort in this dataset corresponds to a single upload batch: 15 of 16 cohorts have exactly one upload_date; one ('AI Connected band Not Interested') spans two adjacent dates. Cohort and 'upload batch' are therefore effectively interchangeable here.

- candidate_phone is missing on exactly 1 row (kept in all aggregates; excluded only from phone-uniqueness checks).
- 17,097 unique candidate_phone values across 18,197 non-blank rows: 1,100 rows share a phone with another row. In every case the repeat is across a DIFFERENT cohort/date, never a true duplicate within the same cohort, see Section 4.

Field definitions used throughout this report (as found in the data):

- Lead = one uploaded row. Attempted = a call attempt was made at least once (0/1).
- Connected = the call connected with the candidate at least once (0/1). tag_filled = the agent filled a call-outcome tag after connecting (0/1).
- Interested = candidate was tagged Interested on a call (0/1).
- OB_after_upload / OB_after_first_attempt = candidate onboarded, measured from upload date vs. from first call attempt.
- FT_after_upload / FT_after_first_attempt = candidate went First Trip (the business outcome), measured from upload date vs. from first attempt.

4. Data Quality & Assumptions

Every issue below was investigated, not silently cleaned. The full live-formula version is in the Excel workbook's Data_Quality tab; SQL Query 9 reproduces the same checks.

Issue	Evidence	Treatment & Reason
Missing candidate_phone	1 row blank (PreOb-Ob Fees Paid, set 1, 29-Jul)	Kept in all cohort/funnel aggregates (every funnel flag is valid); excluded only from phone-uniqueness checks.
Duplicate candidate_phone across cohorts	1,100 rows share a phone with another row; always a DIFFERENT cohort/date (0 same-cohort duplicates)	Not deduplicated a re-targeted candidate is a new, independent lead/business event. Grain is lead-per-cohort, not person.
'Interested' does not gate OB/FT	116 of 119 OB events and 53 of 54 FT events have Interested = 0	Interested reported as a process/tagging metric only, never as a required funnel checkpoint; flagged wherever it could mislead.
Right-censored / immature cohorts	Cohorts uploaded in the last few days show 0-low FT, but FT elsewhere takes 1-2+ weeks to register	Top-3 ranking and cross-cohort comparisons restricted to cohorts aged ≥5 days since upload.
Tiny cohorts	5 of 16 cohorts have <5 leads (as low as 1)	Excluded from ranking via a leads ≥1,000 minimum-size filter; kept visible in the full cohort table.
Zero-denominator risk	Connect/interest rates undefined when Attempted/Connected = 0	All rate formulas wrapped in IFERROR / NULLIF / SAFE_DIVIDE, returning 'n/a' instead of an error or a misleading 0%.
FT_after_first_attempt < FT_after_upload	Every FT_after_first_attempt = 1 row also has FT_after_upload = 1 (0 exceptions); same for OB	Confirms these are nested definitions (narrower vs. wider window), not independent outcomes; FT_after_upload used as the primary metric since it is the more complete count.

Key analytical assumption: 'today' for cohort-age / maturity purposes is taken as the latest upload_date present in the file (6 Aug 2026), since no separate data-extraction timestamp is given. This is stated explicitly because it directly affects which cohorts are judged 'mature enough' to rank.

5. Funnel Analysis

Overall funnel, all 16 cohorts combined:

Stage	Count	% of Previous Stage	% of Total Leads
Lead	18,198	-	100.00%
Attempted	11,973	65.79%	65.79%
Connected	5,550	46.35%	30.50%
Interested	348	6.27%	1.91%
OB_after_upload	119	34.20% *	0.65%
FT_after_upload	54	45.38%	0.30%

*Caution: the 'OB / Interested' figure overstates the true relationship 116 of the 119 OB events happen WITHOUT Interested ever being set to 1 (Section 4). The '% of Total Leads' column is the reliable, path-independent view of every stage and is what the drop-off analysis below is based on.

Largest drop-offs (leads lost, vs. total leads)

Transition	Leads Lost	% of Total Leads Lost
Lead → Attempted (never called)	6,225	34.21%
Attempted → Connected (no answer)	6,423	35.30%
Connected → Interested	5,202	28.59%
Interested-stage → OB	229	1.26%
OB → FT	65	0.36%

Reading: the two largest leaks by volume are Lead→Attempted and Attempted→Connected, together accounting for roughly 70% of all leads. Lead→Attempted (34.2% of all leads never called) is the more actionable of the two, since it is entirely within operational control no candidate behavior is involved, only call-center capacity and prioritization. Attempted→Connected depends partly on candidate availability, so is comparatively harder to move. Connected→Interested is also large in absolute count but is complicated by the Interested-tagging inconsistency documented in Section 4: some of this 'drop' is agents not applying the tag rather than candidates genuinely disengaging.

Which of these are causal vs. descriptive: the funnel counts themselves are purely descriptive they say where volume is lost, not why. Whether calling a lead sooner, or calling it more times, actually increases connect/FT probability cannot be established from this data alone, since call timing/count is not randomly assigned (agents likely prioritize leads they judge as promising, which would bias any naive before/after comparison). Any operational change based on this funnel should be validated with an A/B test (Section 10) before being treated as causal.

6. Cohort Analysis & Top Performing Cohorts

Metric chosen: Lead → FT_after_upload conversion rate. This is the end business outcome (not an intermediate proxy like Interest or Connect rate), and FT_after_upload is the more complete of the two FT fields (54 events vs. 17 for FT_after_first_attempt).

Filters applied, and why:

- Minimum size: leads ≥ 1,000. Below this, a conversion rate is a single-event artifact e.g. a cohort with 1 lead and 0 FT is not meaningfully '0%'; it is unmeasured. This threshold was chosen because it is the point at which the 11 largest cohorts (of 16) fall, cleanly separating them from 5 near-zero-volume cohorts (1-3 leads each).

- Minimum maturity: age ≥ 5 days since upload. FT events in this data take well over a week to register even in cohorts that eventually convert well (e.g. Khanna-2W, uploaded 10 days before the data's latest date, and Single Referral, 12 days). Comparing a 0-day-old cohort's 0% FT rate to a 12-day-old cohort's 0.9% rate would penalize the newer cohort purely for not having had time to convert yet (right-censoring).

Top 3 cohorts

Rank	Cohort	Leads	Age (days)	FT	Lead→FT %
1	Single Referral > 7 days - 24th Jul	1,500	12	14	0.933%
2	Khanna - 2W 26th Jul	1,546	10	14	0.906%
3	PreOb-Ob Fees Paid 29th Jul (set 1)	1,483	8	7	0.472%

Is the top cohort genuinely strong, or just small-sample noise? At 14/1,500 vs. 14/1,546, ranks 1 and 2 are close enough that their relative order should not be over-read with only 14 events each, a couple of leads swinging the outcome either way could flip the ranking.

What strengthens confidence in the top group as a whole:

(a) PreOb-Ob Fees Paid was uploaded as two independent batches (set 1 and set 2) on consecutive days, and both land in a similar, above-average range (0.47% and 0.45% respectively) a real campaign effect should reproduce across batches, and it does;

(b) the driver model in Section 8 independently assigns Single Referral and Khanna-2W positive, cohort-specific coefficients even after controlling for Attempted/Connected/Interested, meaning the effect is not simply explained by those cohorts getting called more.

Interpretation of the top cohorts

- Single Referral > 7 days - 24th Jul: highest raw and model-adjusted FT rate, despite a very low Interested-tag rate (1.0%) most of its conversions bypass that tag entirely, consistent with referral-sourced candidates arriving with higher pre-existing intent that a phone-tag doesn't capture.
- Khanna - 2W 26th Jul: second-highest FT rate with a much higher attempt rate (89%) than Single Referral (97% attempted also, similar) strong operational execution on a cohort that also converts well.
- PreOb-Ob Fees Paid (sets 1 & 2): distinctly higher Interested-tag rate (16%, vs. ~1-5% elsewhere) these are pre-qualified/fee-paid candidates, so agents may be tagging more conservatively-but-accurately here. FT rate is solid but roughly half of ranks 1-2, worth watching as volume scales.

Caveat common to all three: base rates are low (7-14 FT events each), so all reported percentages carry real sampling uncertainty; a small operational disruption (agent turnover, a bad calling day) could visibly move next-period numbers even with no change in underlying lead quality.

7. Aggregation / SQL Logic

The equivalent analysis was written as 11 commented BigQuery Standard SQL queries each with a stated business question, logic, and output, using CTEs and SAFE_DIVIDE/NULLIF to guard every division. Highlights:

- Query 4 (cohort aggregation) groups by lead_source and reproduces the Cohort_Analysis tab exactly.
- Query 5 (top-3 cohorts) applies the same leads $\geq 1,000$ AND age ≥ 5 days filters used in Excel, so the ranking is not tool-dependent.
- Query 8 (Interested-flag leakage check) and Query 9 (funnel-state hierarchy validation) reproduce the Section 4 data-quality findings directly against the raw table.
- Query 10 is a sanity check that SUMs of the cohort-level table reconcile exactly to the single-row overall funnel required before publishing any number in this report.

Dialect note: BigQuery Standard SQL was used per the brief's default; SAFE_DIVIDE replaces the NULLIF-guarded division used in Excel, achieving the same zero-denominator protection.

8. Conversion Driver / Model Analysis

What the case means by 'model': factors that influence the chance of FT, with a confusion matrix. Given the extreme rarity of FT (54 of 18,197 leads, 0.30%) and the intern's stated Excel/SQL strength over Python, the model was built to be genuinely interpretable and defensible, not a black box a logistic regression, whose coefficients can be read directly as 'this factor raises or lowers the odds of FT'.

Setup

- Target: FT_after_upload (0/1).
- Features: lead_source (cohort, one-hot encoded), Attempted, Connected, tag_filled, Interested, Attempt-per-Lead, upload-to-first-attempt speed (hours).
- Deliberately excluded (leakage): OB_after_upload and OB_after_first_attempt OB is effectively the funnel stage immediately before FT; including it would let the model 'read off' a near-final-stage outcome instead of finding genuine upstream drivers. FT_after_first_attempt was also excluded as a narrower version of the target itself.
- Evaluation: 5-fold stratified cross-validation with class_weight = 'balanced' to counter the 0.3% base rate; every prediction reported is out-of-fold.

Confusion matrix (out-of-fold, default 0.5 threshold)

	Predicted: No FT	Predicted: FT
Actual: No FT	11,717	6,426
Actual: FT	11	43

Metric	Value	Interpretation
Precision (FT class)	0.7%	Very low expected at a 0.3% base rate; a class-balanced model flags ~35% of all leads to catch most true FTs, so most flags are false positives at this threshold.
Recall (FT class)	79.6%	The model catches ~4 of every 5 leads that actually go FT.
ROC-AUC	0.798	Good ranking ability leads scored higher by the model are meaningfully more likely to convert, even though a hard cutoff is not operationally usable.

Recommended use: rank leads by predicted probability to prioritize call-center capacity toward the highest-propensity leads (a scoring/triage tool), rather than using the model as a hard accept/reject classifier. At 54 total positive events, precision at any fixed threshold will remain low regardless of model choice the AUC/ranking signal is the trustworthy part of this output, not the confusion-matrix cell counts in isolation.

Top drivers

Feature	Coefficient	Direction	Read
Attempted	5.22	Strong positive	Being called at all is the single strongest driver consistent with Section 5's finding that 34% of leads are never attempted.
tag_filled	2.63	Strong positive	A completed call-outcome tag correlates with much higher FT odds likely reflects agent engagement/call quality, not an independently addable lever.
Cohort: Khanna-2W / Single Referral	2.25 / 2.01	Positive	Matches their #1-2 ranking by raw Lead→FT rate a genuine cohort effect, not explained away by call volume alone.
Cohort: PreOb-Ob Fees Paid (sets 1&2)	1.52 / 1.27	Positive	Both independently-uploaded batches show a similar, positive effect reduces the chance this is single-batch noise.
Attempt-per-Lead	0.46	Weak positive	More attempts associates with slightly higher odds, a small effect once 'Attempted' is already in the model.
Connected, Interested	-2.03, -2.05	Negative (caution)	Counter-intuitive; most likely a statistical suppressor effect from collinearity with cohort/Attempted, and from Section 4's finding that most FT/OB events bypass the normal Connected→Interested path. Not a signal to avoid connecting or tagging candidates.
Cohort: 2W3W batches / AI Connected band Not Interested	-3.5 to -4.3	Negative	These cohorts recorded zero FT despite being mature (≥19 days) a genuine underperformance signal, not a censoring artifact.

Correlation vs. causation, explicitly: every coefficient above describes association within this observational dataset, not a causal effect. Agents were not randomly assigned which leads to call, when, or how many times they likely prioritize leads they already judge as more promising, which can inflate the apparent effect of 'being attempted' or 'being tagged.' None of these relationships should be reported to stakeholders as 'X causes more FT' without a controlled experiment (Section 11).

9. Key Findings

Finding 1: Over a third of leads are never called

Evidence: 6,225 of 18,198 leads (34.2%) have Attempted = 0.

Hypothesis: call-center capacity or lead-prioritization logic is not covering the full upload volume in time.

Recommended action: audit dialer/CRM routing to confirm every uploaded lead enters a call queue; if capacity-constrained, prioritize the highest-propensity leads first using the driver-model score (Section 8).

Experiment: A/B test route a random half of new uploads through a 'guaranteed first attempt within 24h' queue vs. business-as-usual; compare attempt-rate and downstream FT rate.

Success metric: Attempt rate (target: >90%), tracked weekly by cohort.

Finding 2: 'Interested' is not a reliable funnel gate

Evidence: 116 of 119 OB events and 53 of 54 FT events occur without Interested ever being set to 1; the tag's usage rate varies from <1% to ~16% across cohorts with no clear link to actual FT rate.

Hypothesis: agents apply the Interested tag inconsistently (some skip it after a positive call, some use it as a formal qualification step, some cohorts' scripts don't prompt for it).

Recommended action: stop using Interested as a required funnel stage in dashboards/incentive structures; retrain agents on consistent tagging if the business wants it to become a leading indicator, or replace it with a hard system event (e.g. a scheduling action) as the qualification signal.

Experiment: standardize the call script/tagging prompt in one cohort for 2 weeks and compare Interested-tag consistency (and its correlation with FT) against untouched cohorts.

Success metric: Interested-tag rate variance across cohorts (target: narrow the current <1%-16% spread).

Finding 3: Certain cohorts convert meaningfully better, and it survives model controls

Evidence: Single Referral, Khanna-2W, and PreOb-Ob Fees Paid (both batches) rank highest on raw Lead→FT rate AND retain positive, cohort-specific coefficients in the driver model after controlling for Attempted/Connected/Interested.

Hypothesis: these sourcing channels bring higher pre-existing intent or better-qualified candidates than others (e.g. referral-based sourcing, fee-paid pre-qualification).

Recommended action: shift incremental lead-sourcing budget/volume toward these channel types where feasible; deprioritize channels showing zero FT at maturity (2W3W batches, 'AI Connected band Not Interested').

Experiment: increase Single-Referral-style sourcing volume by a fixed increment next cycle and compare FT rate to the current baseline, holding call-center capacity constant.

Success metric: Lead→FT rate by cohort type, tracked cohort-over-cohort.

Finding 4: The Attempted→Connected step loses more volume than any stage except the missed-attempt gap

Evidence: only 46.4% of attempted leads connect (5,550 of 11,973); this is 35.3% of all leads lost at this single step.

Hypothesis: call timing (time-of-day, days since upload) may not be optimized for candidate availability.

Recommended action: analyze connect rate by call time-of-day and by upload-to-first-attempt speed bucket (SQL Query 7) to identify a better calling window, if the underlying call-timestamp data (not present in this extract) can be obtained.

Experiment: retry non-connected leads at a different time-of-day than the first attempt and measure incremental connect rate.

Success metric: Connect rate (of attempted), tracked weekly.

10. Product Recommendations

Prioritized by potential impact, confidence in evidence, and implementation effort:

Priority	Recommendation	Impact	Confidence	Effort
1	Close the never-attempted gap (34% of leads) audit routing / capacity so every lead gets at least one call	High largest absolute volume in the funnel	High direct count, not inferred	Low-Medium operational/CRM fix
2	Shift sourcing mix toward Single-Referral / Khanna-2W-style channels; deprioritize zero-FT mature cohorts (2W3W batches)	Medium-High compounds over future uploads	Medium small event counts, but model-corroborated	Medium sourcing/vendor decision
3	Stop treating 'Interested' as a funnel gate in dashboards/incentives; standardize tagging if it's to be kept as a signal	Medium improves measurement integrity, indirect FT impact	High directly observed in the data	Low reporting/process change
4	Investigate call-timing / connect-rate optimization for the Attempted→Connected step	Medium second-largest volume leak	Low needs call-timestamp data not in this extract	Medium needs additional data + experiment
5	Use the driver-model score to triage/prioritize call-center capacity toward high-propensity leads	Medium improves efficiency of existing capacity	Medium AUC 0.80 but low precision at any fixed cutoff	Medium requires scoring pipeline + monitoring

11. Limitations

- Extreme class imbalance: only 54 FT events total. Every cohort and model statistic in this report carries meaningful sampling uncertainty; small operational changes could visibly move next-period numbers without any real change in underlying quality.
- Right-censoring: cohorts uploaded in the final days of the extract have not had full time to convert; the maturity filter mitigates but does not eliminate this even the 'mature' cohorts (12–19 days old) may still be under-counting FT that occurs later.
- No cost/revenue data: cohorts are ranked purely on conversion rate, not cost-per-lead or cost-per-FT, since no spend field exists in this dataset. A cohort with a lower FT rate but much lower cost per lead could still be more efficient this report cannot assess that.
- No call-timestamp or agent-identity data: the Attempted→Connected drop-off cannot be decomposed by time-of-day or by agent, limiting how actionable Finding 4 can be made without additional data.
- Observational, not experimental data: every driver identified (attempt behavior, tag_filled, cohort) is a correlation, not a proven causal lever see Section 8. Recommendations 1-3 in Section 10 are proposed as testable hypotheses, not settled conclusions.
- Single-snapshot extract: this analysis reflects the data as of the latest upload_date in the file; FT and OB counts for recent cohorts will continue to rise as more time passes, and this report will understate their eventual conversion rate.